HX-400 series centrifugal pumps: operation and maintenance

Operation and Maintenance Manual OM-HX4, edition 3 · Halvorsen Fluid Systems

Manual OM-HX4, edition 3
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Covers HX-400, HX-420 and HX-460, all serial numbers
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1 About this manual

This manual covers installation, commissioning, routine maintenance and fault finding for the three pumps of the HX-400 series. It is written for the operator's maintenance staff and for Halvorsen field engineers. Keep it with the pump and read the safety section before doing any work on the unit.

Technical notes issued after this edition may change individual values, such as alarm thresholds or service intervals. Where a technical note and this manual disagree, the technical note applies. The current list of technical notes for the series is available from Halvorsen Field Service.

2 Safety

A centrifugal pump stores energy in its rotating parts, in the pressurised liquid and, on hot duties, in the heat of the casing. Before any maintenance, isolate the motor electrically and lock the isolator in the off position, close the suction and discharge valves, and let the casing drain and cool. Never rely on the controller's stop command as isolation. On hot liquor and steam-cleaning duties the casing can stay above 60 °C for an hour after the pump has stopped.

Do not run the pump dry, even for a short test. The mechanical seal depends on the pumped liquid for cooling and can be destroyed within seconds without it. Do not run the pump against a closed discharge valve for more than one minute, because the liquid in the casing heats up quickly and can flash to steam.

3 Specifications

The three models share the same bearing frame, shaft seal and baseplate dimensions, so most spare parts are common to the whole series. They differ in impeller diameter, motor power and shaft size.

Table 3-1: Rated performance at 2,900 rpm with water at 20 °C

Model	Hydraulic		Drive	Mechanical	
	Rated flow	Rated head	Motor power	Impeller diameter	Weight with motor
HX-400	60 m³/h	32 m	11 kW	176 mm	148 kg
HX-420	60 m³/h	48 m	15 kW	212 mm	172 kg
HX-460	110 m³/h	30 m	18.5 kW	184 mm	196 kg

The maximum allowed liquid temperature is 110 °C for all models with the standard seal and 140 °C with the high-temperature seal kit HXS-HT. The maximum casing pressure is 16 bar. The pumps are suitable for clean liquids with a viscosity up to 50 mm²/s and a solids content below 0.1 percent by volume.

4 Installation

4.1 Foundation and baseplate

Mount the pump on a rigid concrete foundation at least one and a half times the weight of the complete unit. Level the baseplate with shims under the anchor bolts, then fill the space under the baseplate with non-shrinking grout and let it cure for at least 48 hours before tightening the anchor bolts to their final torque. A baseplate that is not fully grouted will vibrate and shorten bearing and seal life.

4.2 Pipework

Support the suction and discharge pipes independently so that no pipe load is carried by the pump casing. Keep the suction line as short and straight as possible, with an eccentric reducer flat side up in front of the pump inlet to avoid air pockets. A straight run of at least five pipe diameters directly upstream of the inlet is recommended.

4.3 Shaft alignment

The pump and motor are aligned at the factory, but the alignment always changes during transport and grouting. Check it again after the pipework is connected and correct it by moving the motor, never the pump. Use a laser or dial-gauge alignment tool and work to the tolerances below.

Table 4-1: Coupling alignment tolerances

Check	Up to 1,500 rpm		Above 1,500 rpm	
	Acceptable	Target	Acceptable	Target
Parallel offset	0.15 mm	0.08 mm	0.10 mm	0.05 mm
Angular misalignment, per 100 mm	0.12 mm	0.06 mm	0.08 mm	0.04 mm
Coupling gap	3.0 to 4.0 mm, target 3.5 mm, at any speed			

5 Commissioning

Before the first start, fill the pump and suction line completely with liquid and vent the casing through the vent plug on top of the volute. Turn the shaft by hand to check that it rotates freely. Jog the motor briefly to confirm the direction of rotation, which must match the arrow cast on the bearing frame. Running in reverse does not damage the pump immediately but greatly reduces flow and head.

Start the pump with the discharge valve almost closed, then open it slowly until the duty point is reached. During the first hour of operation, check the mechanical seal for leakage, listen for unusual noise and record the drive-end bearing temperature every fifteen minutes. A temperature that is still rising after the first hour points to over-greasing, misalignment or a pipe load on the casing.

6 Routine maintenance

6.1 Maintenance schedule

The intervals below are for continuous duty in a plant room below 30 °C. Shorten them for hot environments, frequent starts or abrasive liquids.

Table 6-1: Routine maintenance intervals

Done by	Task	Interval		Notes
		HX-400, HX-420	HX-460	
Operator	Visual check for leaks and noise	Daily		Walk past with the pump running
	Record bearing temperature and vibration	Weekly		Automatic where telemetry is fitted
Maintenance staff	Inspect mechanical seal	2,000 h		Replace if leakage exceeds a few drops per minute
	Re-grease drive-end bearing	4,000 h	3,000 h	See section 6.2
	Check coupling and alignment	8,000 h		Replace the coupling insert if cracked
	Replace bearings	24,000 h	20,000 h	Or earlier if the temperature trend rises

6.2 Lubrication

The drive-end bearing is re-greased through the grease nipple on top of the bearing housing. Use a lithium-complex grease of consistency class NLGI 2 with a mineral base oil, such as Halvorsen grease HG-2. Add 12 grams per re-greasing on the HX-400 and HX-420 and 15 grams on the HX-460, with the pump running if it is safe to do so. Over-greasing is as harmful as under-greasing: excess grease churns inside the housing and raises the bearing temperature. The non-drive-end bearings are greased for life and need no attention between bearing changes.

6.3 Mechanical seal

A new mechanical seal may weep slightly during the first hours of operation while the faces bed in. After that, any visible leakage means the seal faces are worn or damaged. To replace the seal, remove the motor and coupling, withdraw the back pull-out unit, remove the impeller and slide the seal off the shaft sleeve. Fit the new seal without touching the faces, and lubricate the elastomers with clean water or glycerine, never with oil or grease.

6.4 Impeller removal and refitting

The impeller is held by a nut with a left-hand thread. Lock the shaft with the coupling key before loosening it. When refitting, use a new impeller nut washer and tighten the nut to 95 Nm on the HX-400 and HX-460 and to 110 Nm on the HX-420. Check the impeller clearance to the wear ring afterwards; it should be between 0.3 and 0.5 mm on the diameter.

7 Fault finding

Table 7-1: Common faults, causes and remedies

Symptom	Likely cause	Remedy
Pump delivers no liquid	Casing not filled, or air in the suction line	Fill and vent the casing; check the suction line for leaks
Flow or head too low	Wrong direction of rotation, or worn wear ring	Check the motor wiring; measure the impeller clearance
Noise like gravel in the pump	Cavitation from insufficient suction pressure	Raise the suction level or reduce suction losses
Bearing temperature rising over weeks	Lubricant loss or wrong grease	Check the bearing type and re-grease; see current technical notes
Bearing hot soon after a service	Over-greasing or misalignment	Run for an hour to purge excess grease; recheck alignment
Strong vibration	Misalignment, unbalanced impeller or loose baseplate	Check alignment, impeller condition and grouting

8 Spare parts

Order spare parts through Halvorsen Field Service, quoting the pump model and the serial number from the rating plate. The parts below are the ones most often needed during routine maintenance.

Table 8-1: Recommended spare parts and stock levels

Part	Part number	Used on	Keep in stock	
			1 to 4 pumps	5 pumps or more
Drive-end bearing	BRG-6207-2RS	All models	1	2
Non-drive-end bearing pair	BRG-7306-DB	All models	1	1
Mechanical seal, standard	HXS-35-STD	All models	1	2
Mechanical seal, high temperature	HXS-HT	All models	only for duties above 110 °C	
Wear ring	HXW-176	HX-400	1	1
	HXW-212	HX-420	1	1
	HXW-184	HX-460	1	1
Coupling insert	HXC-28	All models	2	4
Grease, 400 g cartridge	HG-2	All models	2	6